

ML-DSA (FIPS 204)

Lecture 18

SEC 595 — Encrypted Computing

- 1 Sigma Protocols and Fiat–Shamir
- 2 The Scheme
- 3 Security: MLWE and MSIS
- 4 Rejection Sampling
- 5 High Bits and Hints
- 6 Parameters and the NTT

The goal

ML-KEM (Lecture 17) establishes a key. It authenticates nobody.

Today's certificates are signed with RSA or ECDSA. Both fall to Shor.

*ML-DSA is a Fiat–Shamir transformed Sigma protocol
over module lattices, with an abort step.*

This lecture builds that sentence one word at a time.

Definition

$\text{Com}(m; r)$ produces a commitment c ; an opening reveals (m, r) . It is **hiding** if c reveals nothing about m , and **binding** if c cannot be opened to two messages.

A sealed envelope.

Pedersen: in a group of prime order p with generators g, h ,
 $\text{Com}(m; r) = g^m h^r$. A collision with $m \neq m'$ gives
 $\log_g h = (m - m')(r' - r)^{-1} \bmod p$, so binding rests on discrete log.

Definition

Three moves for a relation $\mathcal{R}$: the prover sends a **commitment** a ; the verifier replies with a random **challenge** c ; the prover sends a **response** z .

- 1 **Completeness**: an honest prover always convinces.
- 2 **Special soundness**: from accepting (a, c, z) and (a, c', z') with $c \neq c'$ one extracts a witness.
- 3 **Honest-verifier zero-knowledge (HVZK)**: a simulator with no witness outputs accepting (a, c, z) with the right distribution.

By 2, a cheater answers at most one challenge: $\text{success} \leq 1/|\mathcal{C}|$.

The simulator in 3 works out of order — z first, then a .

Statement $h = g^x$ in a group of prime order p ; witness x .

Protocol

- 1 Commit: random k , send $a = g^k$.
- 2 Challenge: random $c \in \mathbb{Z}_p$.
- 3 Respond: $z = k + cx \bmod p$.

Accept iff $g^z = ah^c$.

Extraction: $x = (z - z')(c - c')^{-1} \bmod p$.

Simulation: pick c, z , set $a = g^z h^{-c}$.

Example — Schnorr, both directions

$p = 23$, $g = 2$ (order 11), $h = 13$, $x = 7$; exponents mod 11.

- a. With $k = 3$, produce accepting transcripts at $c = 2$ and $c = 5$, checking both verification equations.
- b. Extract x from those two transcripts.
- c. Run the simulator at $c = 5$ with $z = 9$: compute a and check the verification equation. Which value was chosen first?

Random oracle model

A hash H is modelled as a truly random function, reachable only by queries.

Fiat–Shamir transform

Replace the verifier's challenge by $c = H(a \parallel \text{ctx})$, where ctx holds the statement and the message. The pair (a, z) — or (c, z) — is a **signature**.

Two rules: the statement must be inside the hash, and the challenge space must be too large to guess before committing.

Example — Fiat–Shamir applied to Schnorr

Same p, g, h, x , with the toy hash $H(a \parallel m) = (3a + 5m) \bmod 11$.

- a. Sign $m = 3$ with nonce $k = 3$: compute a, c, z .
- b. Verify: recompute $a' = g^z h^{-c}$ and check $H(a' \parallel m) = c$.
- c. Does the same signature verify on $m = 4$?

Ring

$$n = 256, \quad q = 8380417 = 2^{23} - 2^{13} + 1, \quad R_q = \mathbb{Z}_q[X]/(X^{256} + 1).$$

Same ring shape as ML-KEM; a larger modulus.

Unlike ML-KEM, seven parameters move between sets: (k, ℓ) , η , τ , γ_1 , γ_2 , ω .

$\beta = \tau\eta$ is derived. (Not the BKZ block size of Lecture 17.)

$$\mathbf{A} \in R_q^{k \times \ell}, \quad \mathbf{s}_1 \in R_q^\ell, \quad \mathbf{s}_2 \in R_q^k, \quad \mathbf{t} = \mathbf{A}\mathbf{s}_1 + \mathbf{s}_2.$$

- $\mathbf{A}$ uniform, expanded from a 32-byte seed ρ .
- $\mathbf{s}_1, \mathbf{s}_2$ short: coefficients uniform on $\{-\eta, \dots, \eta\}$, $\eta \in \{2, 4\}$.
- $\mathbf{A}$ is **rectangular**; the noise is uniform on a box, not a centred binomial.

$(\mathbf{A}, \mathbf{t})$ is an MLWE sample (Lecture 6 §2.3); $\mathbf{s}_2$ is part of the secret key.

Commit. Sample $\mathbf{y} \in R_q^\ell$, coefficients uniform on $(-\gamma_1, \gamma_1]$:

$$\mathbf{w} = \mathbf{A}\mathbf{y}, \quad \mathbf{w}_1 = \text{HighBits}(\mathbf{w}).$$

Challenge. B_τ = polynomials with exactly τ coefficients in $\{\pm 1\}$, the rest 0:

$$\tilde{c} = H(\mu \parallel \mathbf{w}_1) \in \{0, 1\}^{2^\lambda}, \quad c = \text{SampleInBall}(\tilde{c}) \in B_\tau,$$

with μ a digest of the public key and the message.

$|B_\tau| = \binom{256}{\tau} 2^\tau$ must exceed 2^λ : $\tau = 39, 49, 60$ give $2^{192.8}, 2^{225.3}, 2^{257.0}$.

$$\mathbf{z} = \mathbf{y} + c \mathbf{s}_1 \in R_q^\ell.$$

The signature is

$$\sigma = (\tilde{c}, \mathbf{z}, \mathbf{h}),$$

with $\mathbf{h}$ the hint of Section 5.

$\mathbf{w}_1$ is not transmitted: the verifier recomputes it and checks that hashing it reproduces $\tilde{c}$.

Proposition

$$\mathbf{Az} - c\mathbf{t} = \mathbf{Ay} - c\mathbf{s}_2 = \mathbf{w} - c\mathbf{s}_2.$$

Proof: the term $c\mathbf{As}_1$ appears in both $\mathbf{Az}$ and $c\mathbf{t}$. □

From public data the verifier recovers the secret commitment $\mathbf{w}$, perturbed by $c\mathbf{s}_2$. If that cannot change the high bits,

$$\text{HighBits}(\mathbf{Az} - c\mathbf{t}) = \mathbf{w}_1.$$

How big is the perturbation?

Proposition

For $c \in B_\tau$ and $\|\mathbf{s}\|_\infty \leq \eta$: $\|c \mathbf{s}\|_\infty \leq \tau \eta = \beta$.

Each coefficient of $c \mathbf{s}$ is a sum of τ terms $\pm s_j$. □

The generic bound $\|ab\|_\infty \leq n\|a\|_\infty\|b\|_\infty$ (Lecture 3) gives 512 for ML-DSA-44; the truth is 78. Sparsity buys the factor.

One bound, two jobs: $c \mathbf{s}_2$ is what the verifier must tolerate; $c \mathbf{s}_1$ is what the signer must hide.

	Schnorr	ML-DSA
statement	$h = g^x$	$\mathbf{t} = \mathbf{A}\mathbf{s}_1 + \mathbf{s}_2$
witness	x	$(\mathbf{s}_1, \mathbf{s}_2)$, short
commitment	g^k , k uniform	$\mathbf{w}_1 = \text{HighBits}(\mathbf{A}\mathbf{y})$, $\mathbf{y}$ short
challenge	uniform in $\mathbb{Z}_p$	$H(\mu \parallel \mathbf{w}_1)$, sparse ± 1
response	$k + cx$	$\mathbf{z} = \mathbf{y} + c \mathbf{s}_1$
verification	$g^z = a h^c$	$\text{HighBits}(\mathbf{A}\mathbf{z} - c\mathbf{t}) = \mathbf{w}_1$
abort	none	unless $\ \mathbf{z}\ _\infty < \gamma_1 - \beta$
secrecy	DLog	MLWE
unforgeability	DLog	MSIS

Every row but the abort is a translation.

The public key is an MLWE sample $(\mathbf{A}, \mathbf{A}\mathbf{s}_1 + \mathbf{s}_2)$.

Distinguishing it from uniform, or recovering $\mathbf{s}_1$, breaks MLWE — the same assumption as ML-KEM's public key.

Module-SIS (Lecture 6 §2.4)

Given uniform $\mathbf{A} \in R_q^{k \times \ell}$ and a bound γ , find nonzero short $\mathbf{u}$ with $[\mathbf{A} \mid \mathbf{I}] \mathbf{u} = \mathbf{0}$ and $\|\mathbf{u}\|_\infty \leq \gamma$.

SIS is SVP on the q -ary kernel lattice (Lecture 5 §6).

Two accepting transcripts $(\mathbf{w}_1, c, \mathbf{z})$ and $(\mathbf{w}_1, c', \mathbf{z}')$, $c \neq c'$:

$$\text{HighBits}(\mathbf{A}\mathbf{z} - c\mathbf{t}) = \text{HighBits}(\mathbf{A}\mathbf{z}' - c'\mathbf{t}) = \mathbf{w}_1,$$

so $\mathbf{A}(\mathbf{z} - \mathbf{z}') - (c - c')\mathbf{t}$ is small.

Substituting $\mathbf{t} = \mathbf{A}\mathbf{s}_1 + \mathbf{s}_2$ gives a short nonzero $\mathbf{u}$ with $[\mathbf{A} \mid \mathbf{I}]\mathbf{u} = \mathbf{0}$: an MSIS solution.

The formal proof uses a variant, **SelfTargetMSIS**, that folds the hash in. Both estimates are computed; the category is the smaller.

Write $u_i = (c \mathbf{s}_1)_i$, $|u_i| \leq \beta$, and y_i uniform on $(-\gamma_1, \gamma_1]$. Then

$$z_i = y_i + u_i \text{ is uniform on } (-\gamma_1 + u_i, \gamma_1 + u_i].$$

The shape is fixed. The **position moves with the secret**.

- $z_i = \gamma_1 + \beta$ is possible only if $u_i = \beta$.
- $\mathbb{E}[z_i] = u_i$, linear in $\mathbf{s}_1$ through a known c : enough signatures recover the key.

The original NTRUSign was broken this way.

Rejection condition

Accept $\mathbf{z} = \mathbf{y} + c\mathbf{s}_1$ only if

$$\|\mathbf{z}\|_{\infty} < \gamma_1 - \beta.$$

Otherwise discard $\mathbf{y}$, $\mathbf{w}$, c , $\mathbf{z}$ *entirely* and restart with a fresh $\mathbf{y}$.

Nothing from a rejected attempt may survive: a partial reuse leaks through the correlation between attempts.

This is **Fiat–Shamir with aborts** (Lyubashevsky).

Proposition

Fix $|u| \leq \beta$, let y be uniform on $(-\gamma_1, \gamma_1]$, and $z = y + u$. Conditioned on $|z| < \gamma_1 - \beta$, z is uniform on the $2(\gamma_1 - \beta) - 1$ integers of that interval, independent of u .

Proof: for $|v| < \gamma_1 - \beta$, $y = v - u$ satisfies $|y| < \gamma_1$, so every v has probability $1/(2\gamma_1)$. □

An accepted $\mathbf{z}$ carries **zero** information about $\mathbf{s}_1$ — not negligible, none.

HVZK simulator: sample $\mathbf{z}$ uniformly from $\|\mathbf{z}\|_\infty < \gamma_1 - \beta$, set $\mathbf{w}_1 = \text{HighBits}(\mathbf{A}\mathbf{z} - \mathbf{c}\mathbf{t})$, output $(\mathbf{w}_1, c, \mathbf{z})$. Same shape as Schnorr's: z first.

	Schnorr	ML-DSA
mask lives in shift by the secret repair	the group $\mathbb{Z}_p$ a bijection none needed	a box $\ \mathbf{y}\ _\infty \leq \gamma_1$ moves the box reject and restart

$\mathbf{y}$ cannot be uniform on R_q : an unbounded $\mathbf{z}$ would verify trivially and the MSIS argument would collapse.

Example — an abort trace

$\mathbb{Z}_{257}[X]/(X^4 + 1)$, $k = \ell = 1$, $\eta = 1$, $\tau = 2$, so $\beta = 2$; $\gamma_1 = 8$.

$$\mathbf{A} = (3, 1, 4, 1), \quad c = (1, 0, -1, 0),$$

$$\mathbf{s}_1 = (1, -1, 0, 1), \quad \mathbf{s}_2 = (0, 1, -1, 0).$$

- a. Compute $\mathbf{t}$, then $c \mathbf{s}_1$; check $\|c \mathbf{s}_1\|_\infty \leq \tau \eta$.
- b. Attempt one draws $\mathbf{y} = (7, -6, 5, -3)$. Compute $\mathbf{z}$. Accept or reject?
- c. Attempt two draws $\mathbf{y} = (2, -3, 4, 0)$. Compute $\mathbf{z}$. Accept or reject?

Example — what the rejected z would have leaked

Continue: y_i uniform on the 16 values $(-\gamma_1, \gamma_1] = \{-7, \dots, 8\}$,
 $u_i = (c \mathbf{s}_1)_i$.

- a. For each $u_i \in \{-2, \dots, 2\}$, give the range of $z_i = y_i + u_i$.
- b. The rejected candidate had $z_0 = 8$. Which u_0 does that rule out?
- c. For each u_i , count the y_i that survive $|z_i| \leq 5$, and the range of the surviving z_i .
- d. What does c. prove?

Three rejection tests

- ① $\|\mathbf{z}\|_\infty < \gamma_1 - \beta$ — hides $\mathbf{s}_1$.
- ② $\|\mathbf{r}_0\|_\infty < \gamma_2 - \beta$, where $\mathbf{r}_0 = \text{LowBits}(\mathbf{w} - c\mathbf{s}_2)$ — the verifier's high bits agree, and nothing about $\mathbf{s}_2$ leaks.
- ③ $\|c\mathbf{t}_0\|_\infty < \gamma_2$ and $\mathbf{h}$ has at most ω ones — Section 5.

Treating coefficients as independent,

$$\Pr[\text{accept}] \approx \left(\frac{2(\gamma_1 - \beta) - 1}{2\gamma_1} \right)^{256\ell} \left(\frac{2(\gamma_2 - \beta) - 1}{2\gamma_2} \right)^{256k}.$$

Set	β	$\Pr[\mathbf{z}]$	$\Pr[\mathbf{r}_0]$	Both	Attempts
ML-DSA-44	78	0.541	0.430	0.233	4.30
ML-DSA-65	196	0.619	0.316	0.195	5.12
ML-DSA-87	120	0.662	0.390	0.258	3.87

The low-bits test dominates: $\gamma_2 \ll \gamma_1$.

ML-DSA-65 is slowest, not ML-DSA-87: $\eta = 4$ pushes β to 196.

Variable signing time is safe: the loop count depends on fresh $\mathbf{y}$ and public μ . The test itself must be constant-time.

Two decompositions

Power2Round, $d = 13$

$$r = r_1 2^{13} + r_0, \quad -2^{12} < r_0 \leq 2^{12}. \quad r_1 \text{ takes } 2^{10} \text{ values.}$$

Decompose, with $2\gamma_2 \mid q - 1$

$$r = r_1(2\gamma_2) + r_0, \quad |r_0| \leq \gamma_2; \quad \text{HighBits}(r) = r_1, \text{ LowBits}(r) = r_0. \quad r_1 \text{ takes 44 or 16 values.}$$

Power2Round shrinks the *public key*. Decompose shrinks the *commitment*.

A signature must verify *exactly*, so discarded bits must be reconstructed — that is the hint.

The public key carries only $\mathbf{t}_1$

$\mathbf{t} = \mathbf{t}_1 2^{13} + \mathbf{t}_0$. Public key $(\rho, \mathbf{t}_1)$; $\mathbf{t}_0$ goes in sk .

Set	full $\mathbf{t}$	only $\mathbf{t}_1$
ML-DSA-44	2976	1312
ML-DSA-65	4448	1952
ML-DSA-87	5920	2592

About 56% saved. But the verifier can now only compute

$$\mathbf{A}\mathbf{z} - c\mathbf{t}_1 2^{13} = (\mathbf{w} - c\mathbf{s}_2) + c\mathbf{t}_0,$$

and $\|c\mathbf{t}_0\|_\infty$ can reach $39 \cdot 2^{12} = 159744 > \gamma_2 = 95232$.

Definition

$\mathbf{h} \in \{0, 1\}^{256k}$, with $h_i = 1$ exactly where adding $c \mathbf{t}_0$ changes the high bits of $(\mathbf{w} - c\mathbf{s}_2)_i$.

The signer builds it with `MakeHint`; the verifier corrects with `UseHint`.

After test 3, $\|\mathbf{c}\mathbf{t}_0\|_\infty < \gamma_2$, so a crossing moves the high part by exactly ± 1 , in a direction the verifier can see. One bit suffices.

Encoded as ω index bytes plus k count bytes: $\omega + k$ bytes.

Verification

- 1 Recompute μ from the public key and the message.
- 2 $c = \text{SampleInBall}(\tilde{c})$.
- 3 $\mathbf{w}'_1 = \text{UseHint}(\mathbf{h}, \mathbf{A}\mathbf{z} - c \mathbf{t}_1 2^{13})$.
- 4 Accept iff $\tilde{c} = H(\mu \| \mathbf{w}'_1)$, $\|\mathbf{z}\|_\infty < \gamma_1 - \beta$, and $\mathbf{h}$ has at most ω ones.

No loop: verification is straight-line.

Set	(k, ℓ)	η	τ	β	γ_1	γ_2	ω	λ	pk	sk	σ	Cat.
ML-DSA-44	(4, 4)	2	39	78	2^{17}	$(q-1)/88$	80	128	1312	2560	2420	2
ML-DSA-65	(6, 5)	4	49	196	2^{19}	$(q-1)/32$	55	192	1952	4032	3309	3
ML-DSA-87	(8, 7)	2	60	120	2^{19}	$(q-1)/32$	75	256	2592	4896	4627	5

Sizes in bytes. The name is the shape of **A**. No category-1 set.

k and ℓ tune the MLWE and MSIS estimates separately (core-SVP, Lecture 17) — hence a rectangular **A**.

Proposition

$$pk = 32 + 320k, \quad \sigma = \frac{\lambda}{4} + 32\ell(1 + \log_2 \gamma_1) + \omega + k.$$

t_1 : 256 coefficients of 10 bits = 320 bytes per ring element.

$$\text{ML-DSA-44: } 32 + 4 \cdot 576 + 84 = 2420$$

$$\text{ML-DSA-65: } 48 + 5 \cdot 640 + 61 = 3309$$

$$\text{ML-DSA-87: } 64 + 7 \cdot 640 + 83 = 4627$$

z is 95.2% of an ML-DSA-44 signature: γ_1 is the parameter worth fighting over.

A complete NTT at $q = 8380417$

Modulo $X^{256} + 1$ we need a primitive 512-th root of unity, i.e. $512 \mid q - 1$ (Lecture 3).

$$q - 1 = 8380416 = 2^{13} \cdot 3 \cdot 11 \cdot 31.$$

$\zeta = 1753$: $\zeta^{256} \equiv -1$, $\zeta^{512} \equiv 1$. All eight layers run:

$$R_q \cong \prod_{i=0}^{255} \mathbb{Z}_q[X] / (X - \zeta^{2 \text{brv}(i)+1}) \cong \mathbb{Z}_q^{256}.$$

256 linear factors: multiplication is 256 scalar products. ML-KEM's $q = 3329$ stops at 128 quadratic factors.

Example — the two-adicity comparison

For $q = 3329$ and $q = 8380417$, factor $q - 1$ and determine:

- a. the two-adicity, and whether $256 \mid q - 1$ and $512 \mid q - 1$;
- b. how many layers a negacyclic NTT of length 256 can run;
- c. the degree and number of the resulting factors;
- d. the cost of one pointwise step, in coefficient multiplications;
- e. the largest power-of-two n admitting a complete negacyclic NTT.

Three divisibility conditions on q

$512 \mid q - 1$	the NTT is complete;
$2\gamma_2 \mid q - 1$	Decompose has equal intervals;
$2^{13} \mid q - 1$	Power2Round has a full top interval.

Hence $\gamma_2 = (q - 1)/88$ or $(q - 1)/32$, with $88 = 2^3 \cdot 11$ and $32 = 2^5$.

q had to be about 23 bits anyway, to hold γ_1 and the 13 low bits of $\mathbf{t}$.
Two-adicity 13 costs nothing.

Scheme	Construction	pk	sig	Rests on
Ed25519	Fiat-Shamir	32	64	DLog (Shor)
ML-DSA-44	Fiat-Shamir + aborts	1312	2420	MLWE, MSIS
FN-DSA-512	hash-and-sign trapdoor	897	666	NTRU, SIS
SLH-DSA-128s	hypertree of OTS	32	7856	hash security

FN-DSA is smaller but needs constant-time Gaussian sampling and floating point. SLH-DSA is larger but assumes only a hash (Lecture 16).

ML-DSA is the general-purpose default.

- 1 ML-DSA is Schnorr over a module, made non-interactive by Fiat–Shamir: commit $\mathbf{w}_1 = \text{HighBits}(\mathbf{A}\mathbf{y})$, challenge $c = H(\mu \parallel \mathbf{w}_1)$, respond $\mathbf{z} = \mathbf{y} + c\mathbf{s}_1$.
- 2 Verification rests on $\mathbf{A}\mathbf{z} - c\mathbf{t} = \mathbf{w} - c\mathbf{s}_2$ and $\|c\mathbf{s}\|_\infty \leq \tau\eta = \beta$.
- 3 MLWE hides the key; MSIS prevents forgery.
- 4 Without the abort, $\mathbf{z}$ leaks $\mathbf{s}_1$. Rejecting unless $\|\mathbf{z}\|_\infty < \gamma_1 - \beta$ makes $\mathbf{z}$ independent of the secret — the HVZK simulator. About 4–5 attempts.
- 5 Power2Round halves the public key; the hint repairs verification. $512 \mid q - 1$ gives a complete NTT with 256 linear factors.